

BREAST PATHOLOGY

BY

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MSc CLINICAL CYTOLOGY

OBJECTIVES

- To understand the inflammatory diseases of the breast
- To know the different types of breast tumors
- To know and understand different types of breast malignancies

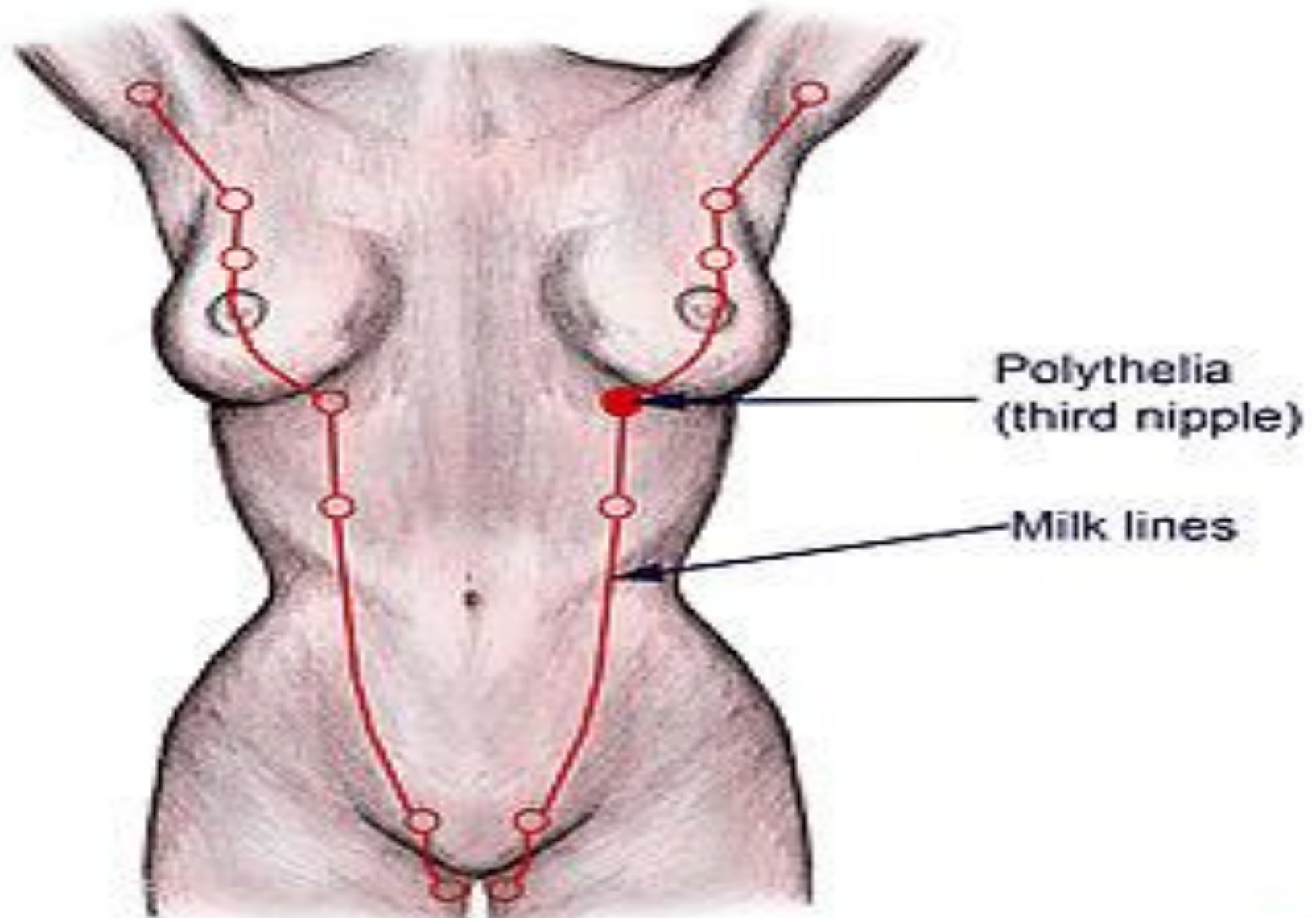
OUTLINE

- Introduction to breast pathology
- Inflammatory conditions
- Benign tumors and fibrocystic changes
- Breast cancer

Modified Sweat Gland

- Embryologically derived from the skin
- Breast tissue (and pathology) can develop anywhere along milk line.

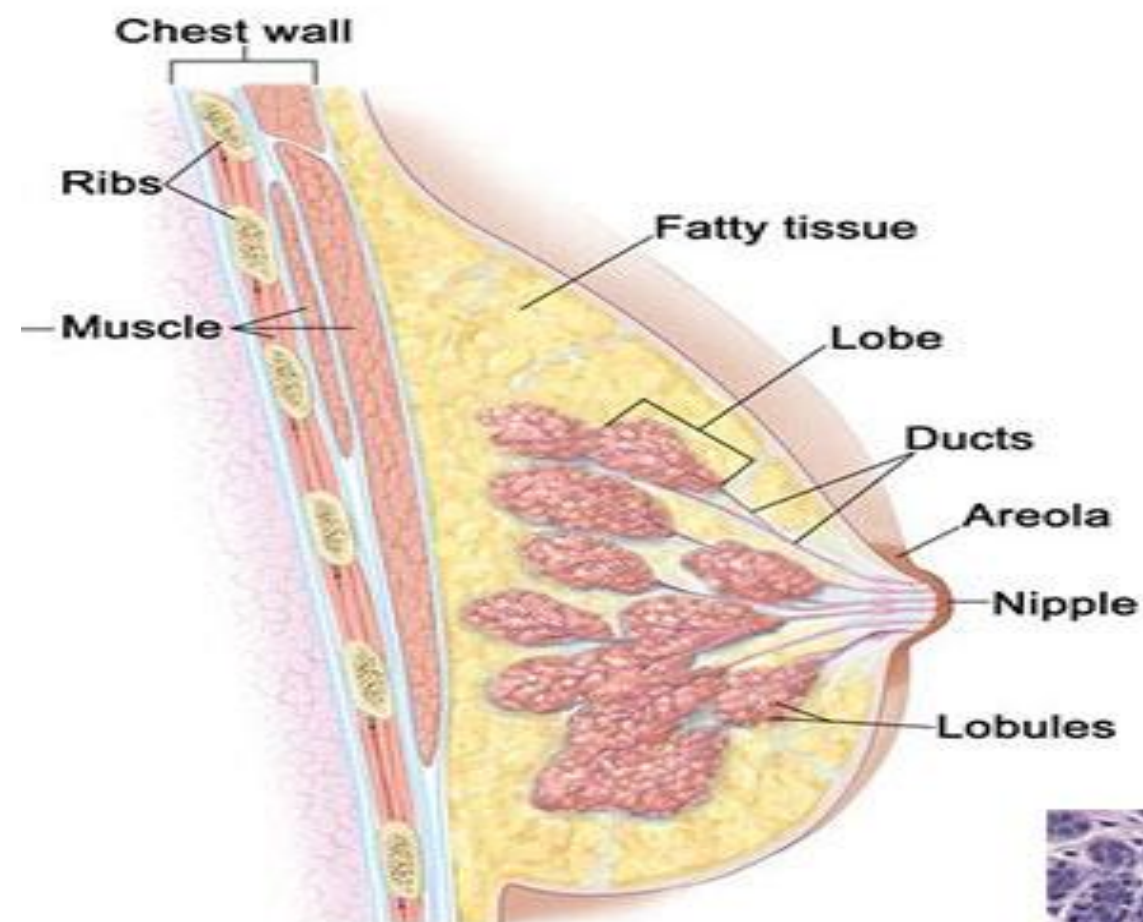
Milk line



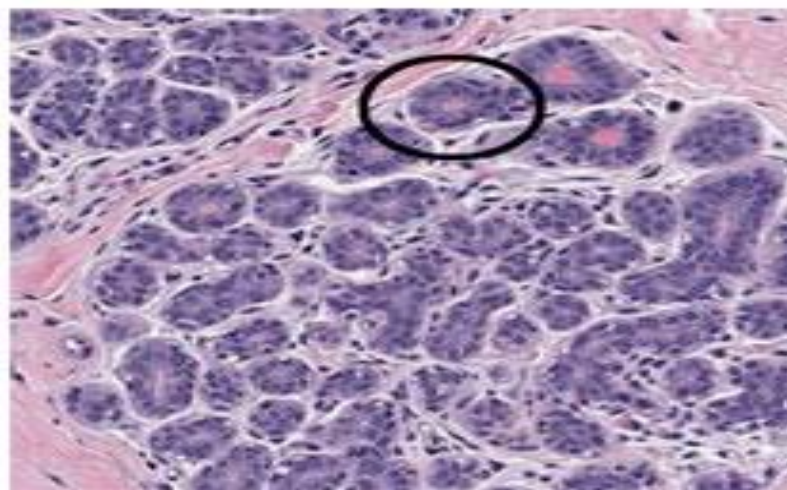
Terminal duct- lobular unit is the functional unit

- Lobules make milk that drains via ducts to nipples

Anatomy of the Female Breast

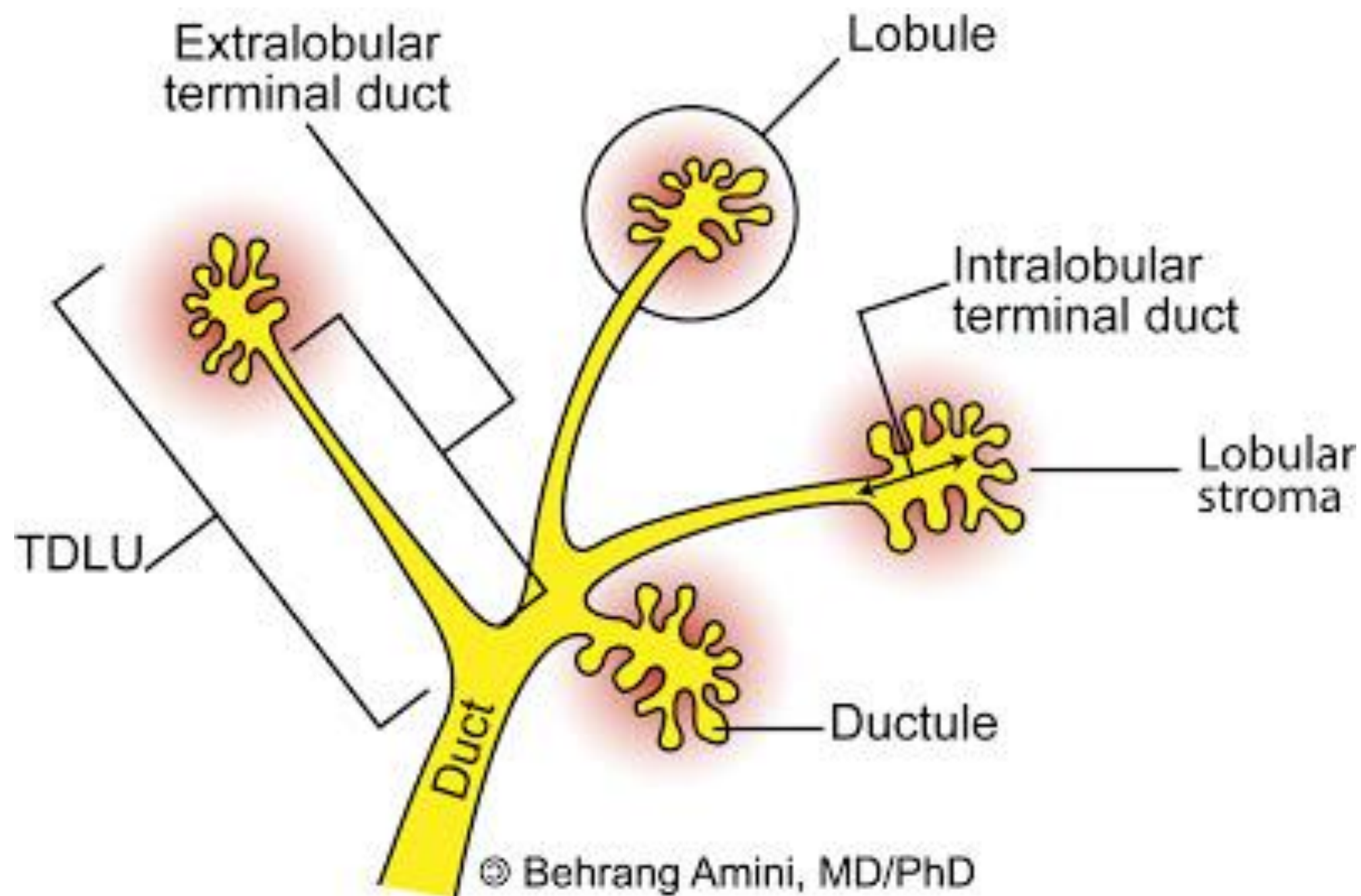


An acinus (circled in black) is an epithelial substructure found in lobules.



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Courtesy of the Susan G. Komen Tissue Bank
at the IU Simon Cancer Center
(www.virtualtissuebank.iu.edu)



Lobules and ducts are lined by two layers of epithelium

- Luminal cell layer
- Myoepithelial cell layer

Hormone sensitive

- Development after menarche
- Breast tenderness during menstrual cycle
- Hyperplasia during pregnancy
- Atrophy after menopause

Galactorrhea

- Milk production outside of lactation
- Not a symptom of breast cancer
- Causes include nipple stimulations, prolactinoma of anterior pituitary and drugs.

INFLAMMATORY CONDITIONS

1. Acute Mastitis
2. Periductal mastitis
3. Ductal ectasia
4. Fat Necrosis

Acute mastitis

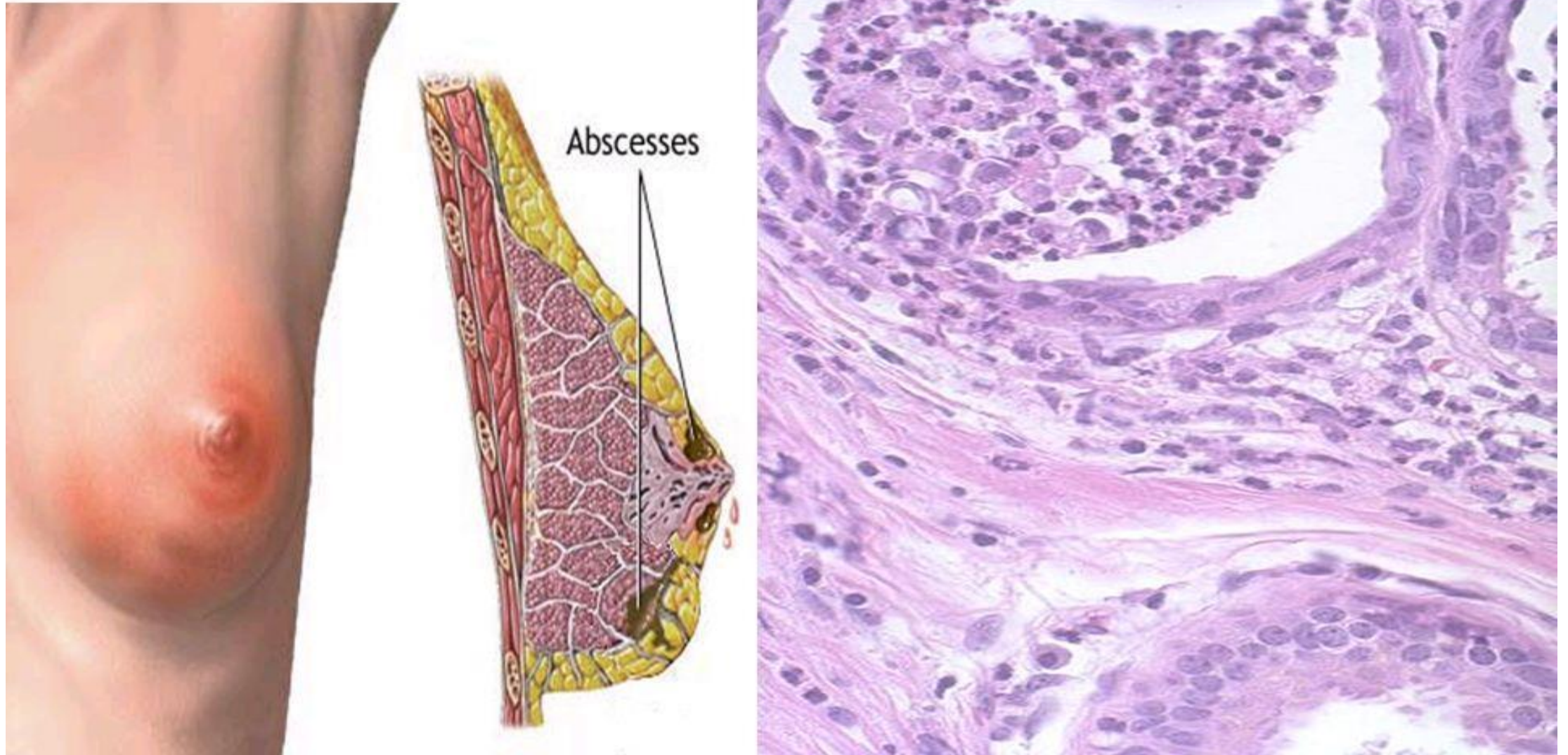
- Bacterial infections of the breast
- Usually due to *S. aureus*
- Associated with breast feeding

Presentation

- Warm, erythematous breast with purulent nipple discharge
- Treatment involves continued drainage and antibiotics. (dicloxacillin)



ACUTE MASTITIS - ABSCESS- BREAST TISSUE IS INFILTRATED BY NEUTROPHILS AND MAY BE NECROTIC



Periductal Mastitis

- Inflammation of the subareolar ducts
- Usually seen in smokers
- Clinically presents as subareolar mass with nipple retraction



Mammary duct ectasia

- Inflammation with dilatation (ectasia) of subareolar ducts
- Rare
- Classically arises in multiparous postmenopausal women

Presentation

- Periareolar mass with green- brown nipple discharge
- Chronic inflammation with plasma cells is seen on biopsy

Fat Necrosis

- Necrosis of breast fat
- Usually related to trauma
- Presents as a mass on exam or calcification on mammography
- Biopsy shows necrotic fat with associated calcifications and giant cells.

